Datasets	Format Examples
Text	<s> KOSMOS-1 can perceive multimodal input, learn in context, and generate output. </s>
Image-Caption	<s> <image> Image Embedding </image> WALL-E giving potted plant to EVE. </s>
Multimodal	<s> <image> Image Embedding </image> This is WALL-E. <image> Image Embedding </image> This is EVE. </s>

Table 21: The examples of the data format to train the KOSMOS-1 model.

C Evaluation

C.1 Input Format Used for Perception-Language Tasks

Figure 7 shows how we conduct zero-shot and few-shot evaluations on perception-language tasks.

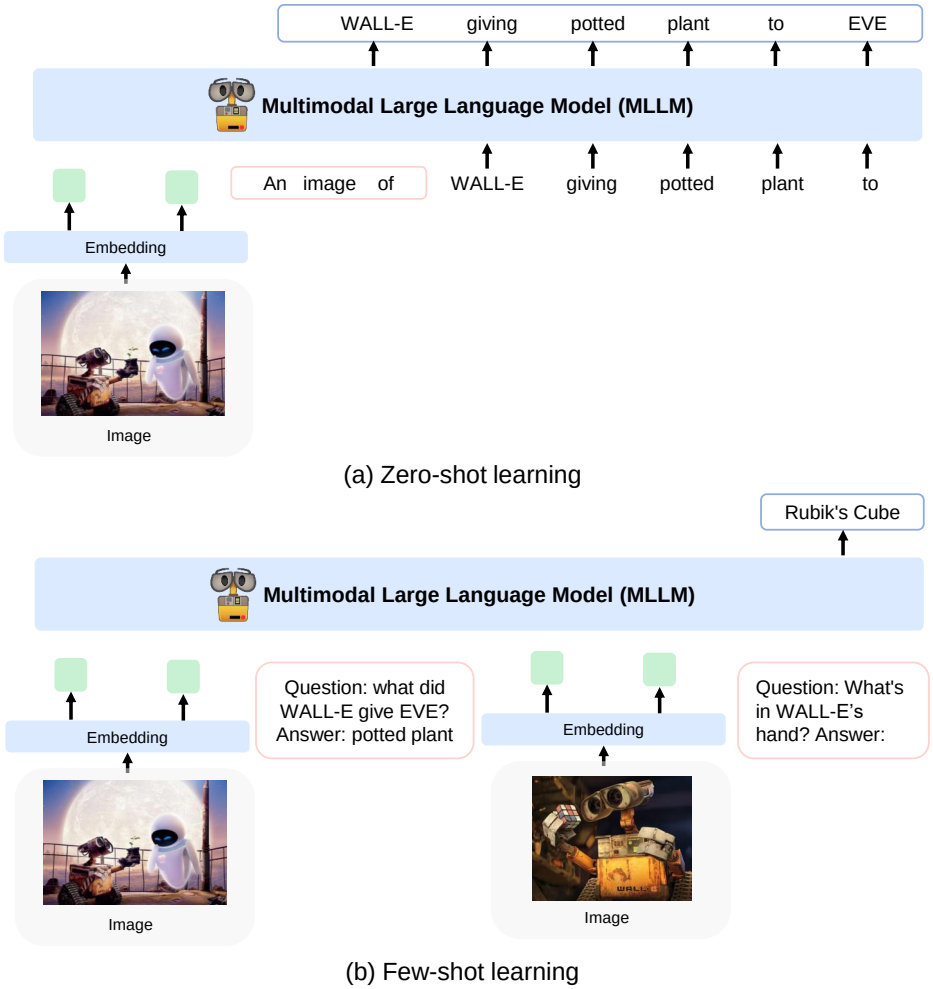


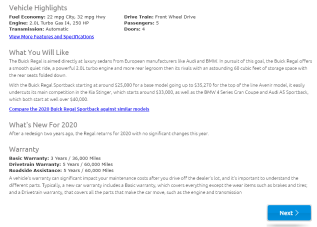
Figure 7: We evaluate KOSMOS-1 on the perception-language tasks in zero- and few-shot settings. (a) Zero-shot learning, e.g., zero-shot image captioning with language prompts. (b) Few-shot learning, e.g., visual question answering with in-context learning.

C.2 Language Tasks

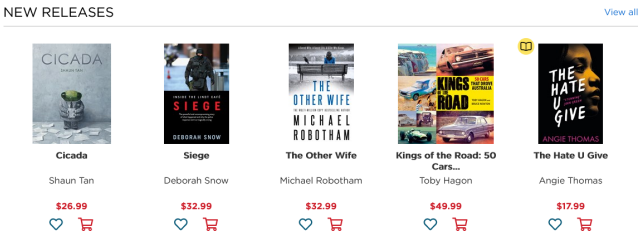
We conduct experiments on language tasks in four categories:

- Cloze and completion tasks: StoryCloze [MRL+17], HellaSwag [ZHB+19]
- Winograd-style tasks: Winograd [LDM12b], Winogrande [SBBC20]
- Commonsense reasoning: PIQA [BZB+20]
- Three datasets from SuperGLUE benchmark [WPN+19]: BoolQ [CLC+19], CB [dMST19], COPA [RBG11]

C.3 WebSRC Task Examples



(a) Question is "*What is the type of this drive?*"



(b) Question is "*Who is the author of "Cicada"?*"

Figure 8: Examples form WebSRC [CZC+21].